Lab 7
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1. Buggy module run with valid and invalid input

```
(base) debangana@Debanganas-Laptop Lab7 % python restaurant_bill_parser.py
==== Running clean_dish_name ====
Enter a dish name from the menu (e.g. *paneer tikka*): paneer tikka
Dish name: Paneer Tikka

==== Running parse_item_prices ====
Enter item prices separated by commas (e.g. 250,180,320): 250,180,320
Item prices: [250, 180, 320]

==== Running split_amount ====
Enter the bill and number of people as TOTAL/PEOPLE (e.g. 1200/4): 1200/4
Each person pays: 300
(base) debangana@Debanganas-Laptop Lab7 %
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(base) debangana@Debanganas-Laptop Lab7 %
(base) debangana@Debanganas-Laptop Lab7 % python restaurant_bill_parser.py
==== Running clean_dish_name ====
Enter a dish name from the menu (e.g. *paneer tikka*): paneer tikka
Dish name: Paneer Tikka

==== Running parse_item_prices ====
Enter item prices separated by commas (e.g. 250,180,320): 780,320,450
Item prices: [780, 320, 450]

==== Running split_amount ====
Enter the bill and number of people as TOTAL/PEOPLE (e.g. 1200/4): 1550/0
Traceback (most recent call last):
  File "/Users/debangana/Software-Engineering/Lab7/restaurant_bill_parser.py", line 40,
in <module>
    main()
    ~~~~^^
  File "/Users/debangana/Software-Engineering/Lab7/restaurant_bill_parser.py", line 37,
in main
    print("Each person pays:", split_amount(value))
                               ~~~~~~~~~~~~^^^^^^^
  File "/Users/debangana/Software-Engineering/Lab7/restaurant_bill_parser.py", line 24,
in split_amount
    return int(total) // int(people)
           ~~~~~~~~~~~^^~~~~~~~~~~~~
ZeroDivisionError: integer division or modulo by zero
(base) debangana@Debanganas-Laptop Lab7 %
```

2. pytest on the buggy module

```
(base) debangana@Debanganas-Laptop Lab7 % python -m pytest test_restaurant_bill_parser.py
============================= test session starts ==============================
platform darwin -- Python 3.13.5, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/debangana/Software-Engineering/Lab7
plugins: hypothesis-6.170.14
collected 3 items

test_restaurant_bill_parser.py FFF                                       [100%]

=================================== FAILURES ===================================
_____________________________ test_clean_dish_name_no_crash _____________________________

s = None

    @given(st.text() | st.none())
    def test_clean_dish_name_no_crash(s):
        try:
>           clean_dish_name(s)

test_restaurant_bill_parser.py:7:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

name = None

    def clean_dish_name(name):
        """
        Removes the chef's-special markers (*, #, ~) and title-cases the name.
        Assumes name is a non-empty string.
        """
>       name = name.strip()
               ^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'

restaurant_bill_parser.py:6: AttributeError

During handling of the above exception, another exception occurred:

    @given(st.text() | st.none())
>   def test_clean_dish_name_no_crash(s):
                ^^^

test_restaurant_bill_parser.py:5:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = None

    @given(st.text() | st.none())
    def test_clean_dish_name_no_crash(s):
        try:
            clean_dish_name(s)
        except Exception as e:
>           assert False, f"clean_dish_name crashed with input {s!r}: {e}"
E           AssertionError: clean_dish_name crashed with input None: 'NoneType' object has no attribute 'strip'
E           assert False
E           Failing test case: test_clean_dish_name_no_crash(
E               s=None,
E           )
E           Explanation:
E               These lines were always and only run by failing test cases:
E                   /Users/debangana/Software-Engineering/Lab7/test_restaurant_bill_parser.py:8

test_restaurant_bill_parser.py:9: AssertionError
_____________________________ test_parse_item_prices_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/debangana/Software-Engineering/Lab7/test_restaurant_bill_parser.py", line 12, in test_parse_item_prices_no_crash
  |     def test_parse_item_prices_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
```

```
|     def test_split_amount_no_crash(s):
|                ^^^
| ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "/Users/debangana/Software-Engineering/Lab7/test_restaurant_bill_parser.py", line 21, in test_split_amount_no_crash
  |     split_amount(s)
  |     ~~~~~~~~~~~~~^^^
  |   File "/Users/debangana/Software-Engineering/Lab7/restaurant_bill_parser.py", line 24, in split_amount
  |     return int(total) // int(people)
  |            ~~~^^^^^^^
  | ValueError: invalid literal for int() with base 10: ''
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "/Users/debangana/Software-Engineering/Lab7/test_restaurant_bill_parser.py", line 23, in test_split_amount_no_crash
  |     assert False, f"split_amount crashed with input {s!r}: {e}"
  | AssertionError: split_amount crashed with input '/': invalid literal for int() with base 10: ''
  | assert False
  | Failing test case: test_split_amount_no_crash(
  |     s='/',
  | )
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "/Users/debangana/Software-Engineering/Lab7/test_restaurant_bill_parser.py", line 21, in test_split_amount_no_crash
  |     split_amount(s)
  |     ~~~~~~~~~~~~~^^^
  |   File "/Users/debangana/Software-Engineering/Lab7/restaurant_bill_parser.py", line 23, in split_amount
  |     total, people = text.split('/')
  |     ^^^^^^^^^^^^
  | ValueError: not enough values to unpack (expected 2, got 1)
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "/Users/debangana/Software-Engineering/Lab7/test_restaurant_bill_parser.py", line 23, in test_split_amount_no_crash
  |     assert False, f"split_amount crashed with input {s!r}: {e}"
  | AssertionError: split_amount crashed with input '': not enough values to unpack (expected 2, got 1)
  | assert False
  | Failing test case: test_split_amount_no_crash(
  |     s='',  # or any other generated value
  | )
  +---------------- 3 ----------------
  | Traceback (most recent call last):
  |   File "/Users/debangana/Software-Engineering/Lab7/test_restaurant_bill_parser.py", line 21, in test_split_amount_no_crash
  |     split_amount(s)
  |     ~~~~~~~~~~~~~^^^
  |   File "/Users/debangana/Software-Engineering/Lab7/restaurant_bill_parser.py", line 23, in split_amount
  |     total, people = text.split('/')
  |                     ^^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'split'
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "/Users/debangana/Software-Engineering/Lab7/test_restaurant_bill_parser.py", line 23, in test_split_amount_no_crash
  |     assert False, f"split_amount crashed with input {s!r}: {e}"
  | AssertionError: split_amount crashed with input None: 'NoneType' object has no attribute 'split'
  | assert False
  | Failing test case: test_split_amount_no_crash(
  |     s=None,  # or any other generated value
  | )
  +------------------------------------
=========================== short test summary info ============================
FAILED test_restaurant_bill_parser.py::test_clean_dish_name_no_crash - AssertionError: clean_dish_name crashed with input None: 'NoneType' object has no a...
FAILED test_restaurant_bill_parser.py::test_parse_item_prices_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_restaurant_bill_parser.py::test_split_amount_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed in 2.59s ===============================
(base) debangana@Debanganas-Laptop Lab7 %
```

3. Fixed module, valid run and bad-input run

```
(base) debangana@Debanganas-Laptop Lab7 % python restaurant_bill_parser.py

==== Running clean_dish_name ====
Enter a dish name from the menu (e.g. *paneer tikka*): paneer tikka
Dish name: Paneer Tikka

==== Running parse_item_prices ====
Enter item prices separated by commas (e.g. 250,180,320): 250,180,320
Item prices: [250, 180, 320]

==== Running split_amount ====
Enter the bill and number of people as TOTAL/PEOPLE (e.g. 1200/4): 1200/4
Each person pays: 300
(base) debangana@Debanganas-Laptop Lab7 %
```

```
(base) debangana@Debanganas-Laptop Lab7 % python restaurant_bill_parser.py

==== Running clean_dish_name ====
Enter a dish name from the menu (e.g. *paneer tikka*): ***
Dish name: Invalid dish name

==== Running parse_item_prices ====
Enter item prices separated by commas (e.g. 250,180,320):
Item prices: Invalid prices

==== Running split_amount ====
Enter the bill and number of people as TOTAL/PEOPLE (e.g. 1200/4): 1200/0
Each person pays: Invalid input, use TOTAL/PEOPLE with people > 0
(base) debangana@Debanganas-Laptop Lab7 %
```

4. pytest on the fixed module

```
(base) debangana@Debanganas-Laptop Lab7 % python -m pytest test_restaurant_bil
l_parser.py
============================= test session starts ==============================
platform darwin -- Python 3.13.5, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/debangana/Software-Engineering/Lab7
plugins: hypothesis-6.170.14
collected 3 items

test_restaurant_bill_parser.py ...                                       [100%]

============================== 3 passed in 0.25s ===============================
```